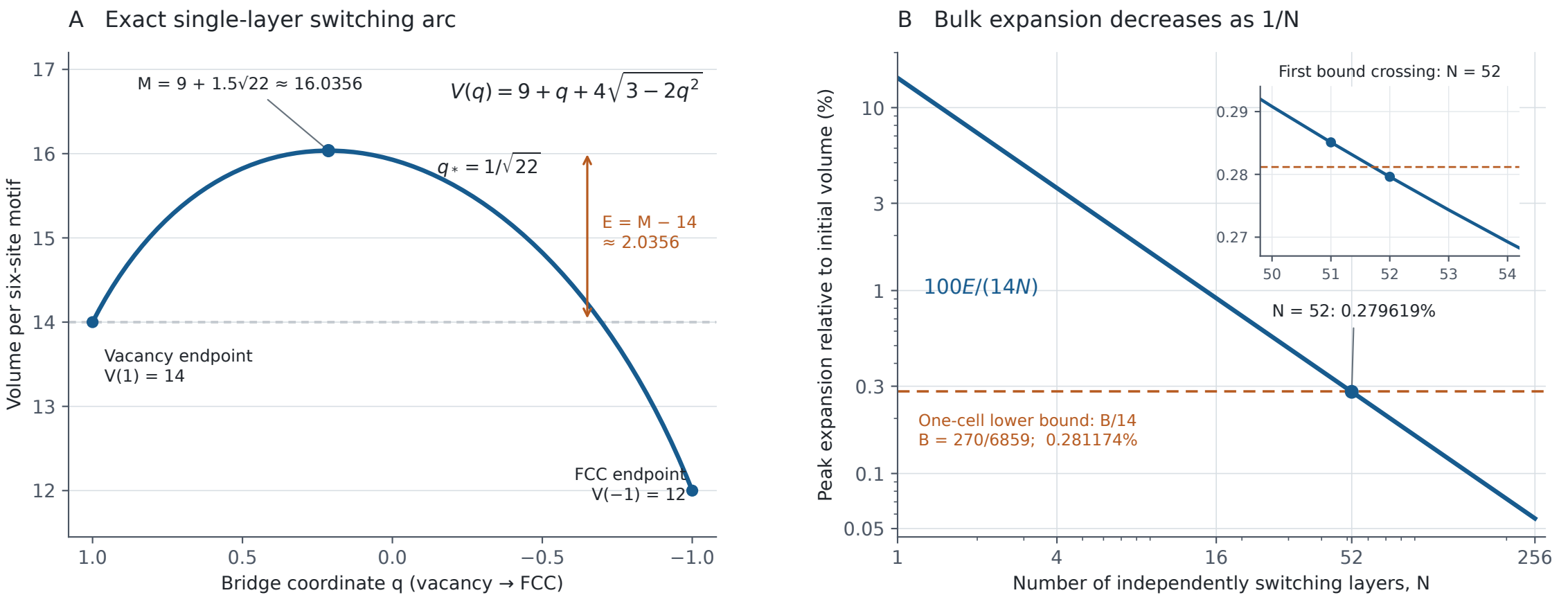

Specified contact-preserving layer family. The N = 52 comparison uses a conservative one-cell lower bound, not its exact optimum.

Kinematic volume bounds; no energy, force, fixed-box transition, or unrestricted supercell optimum is claimed.